1.

Exercise 4

Assume we have the design matrix

$$\boldsymbol{X} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}.$$

Use PCA to reduce the dimension from 2 to 1, i.e., to compute the first principal component.

Solution:

- the *singular value decomposition* (*SVD*) of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times d}$ is

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^\top,$$

where:

  - the columns of $\boldsymbol{U} = [\boldsymbol{u}_1 | \ldots | \boldsymbol{u}_n] \in \mathbb{R}^{n \times n}$, the left singular vectors, are the orthonormal eigenvectors of $\boldsymbol{A}\boldsymbol{A}^\top \in \mathbb{R}^{n \times n}$
  - the columns of $\boldsymbol{V} = [\boldsymbol{v}_1 | \ldots | \boldsymbol{v}_d] \in \mathbb{R}^{d \times d}$, the right singular vectors, are the orthonormal eigenvectors of $\boldsymbol{A}^\top \boldsymbol{A} \in \mathbb{R}^{d \times d}$
  - the diagonal elements of $\boldsymbol{S} = \text{diag}(s_1, \ldots, s_d)$, the singular values of $\boldsymbol{A}$, are such that $s_1^2 \geq \cdots \geq s_d^2$ are the eigenvalues of $\boldsymbol{A}^\top \boldsymbol{A}$

- by taking $\boldsymbol{A} := \frac{1}{\sqrt{n}}\boldsymbol{X}$, we have that:

  - the first l columns of $\boldsymbol{V}$ from the SVD of $\boldsymbol{X}$ are the eigenvectors $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_l$ of $\boldsymbol{\Sigma} = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X}$
  - the corresponding eigenvalues $\lambda_1 \geq \ldots \geq \lambda_l$ are $\frac{s_1^2}{n} \geq \cdots \geq \frac{s_l^2}{n}$, where $s_1, \ldots, s_l$ are the first l singular values of $\boldsymbol{X}$

- thus, $\boldsymbol{w}_k = \boldsymbol{v}_k$ and $\lambda_k = \frac{s_k^2}{n}$, $k = 1, \ldots, l$

$m = 3$, $d = 2$

Lučam $A = \frac{1}{\sqrt{m}} X = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{pmatrix}$

$A^T \cdot A = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$

$\det(A^T \cdot A - \lambda \mathbb{1}) = \begin{vmatrix} \frac{2}{3} - \lambda & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} - \lambda \end{vmatrix} = \left(\frac{2}{3} - \lambda\right)^2 - \left(\frac{1}{3}\right)^2 = 0$

$\left(\frac{2}{3} - \lambda + \frac{1}{3}\right)\left(\frac{2}{3} - \lambda - \frac{1}{3}\right) = 0$

$(1 - \lambda)\left(\frac{1}{3} - \lambda\right) = 0$

$\Rightarrow \lambda_1 = 1 \qquad \lambda_2 = \frac{1}{3}$

$(s_1^2) \qquad (s_2^2)$

- $\lambda_1 = 1$

$(A^T \cdot A - 1 \cdot I_2)\, v_1 = 0 \Rightarrow \begin{pmatrix} \frac{2}{3} - 1 & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$

$\Rightarrow \begin{cases} -\frac{1}{3}a + \frac{1}{3}b = 0 \\ \frac{1}{3}a - \frac{1}{3}b = 0 \end{cases} \Rightarrow a = b$

$\Rightarrow v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$; Normalizăm $m = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\Rightarrow v_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ $w_1 = v_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ $\lambda_1 = \frac{S_h^2}{m} = \frac{1}{3}$

- $\lambda_2 = \frac{1}{3}$

$(A^T \cdot A - \frac{1}{3} I_2)\, v_2 = 0 \Rightarrow \begin{pmatrix} \frac{2}{3} - \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} - \frac{1}{3} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$

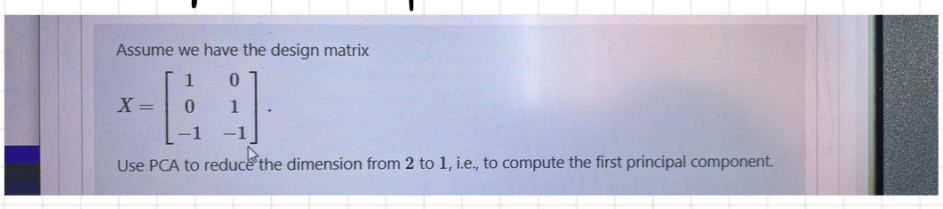
$\begin{cases} \frac{1}{3}a + \frac{1}{3}b = 0 \\ \frac{1}{3}a + \frac{1}{3}b = 0 \end{cases} \Rightarrow a = -b$

$\Rightarrow v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ $m = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

$\Rightarrow v_2 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ $\Rightarrow w_2 = v_2 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ $\lambda_2 = \frac{\frac{1}{3}}{3} = \frac{1}{9}$

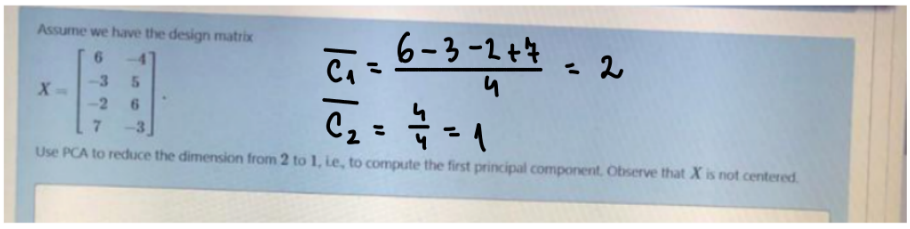
w_1 este prima componentă cu $\lambda_1 = \frac{1}{3}$

La fel



2.

6. Assume we have the design matrix X = [6 -4; -3 5; -2 6; 7 -3]. Use **PCA** to reduce the dimension from 2 to 1 to compute the first principal component. Observe that X is not centered.



$$\Rightarrow \quad \overline{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \qquad n = 4 \quad d = 2$$

$$\Rightarrow \quad X = \begin{pmatrix} 4 & -5 \\ -5 & 4 \\ -4 & 5 \\ 5 & -4 \end{pmatrix} \qquad \Rightarrow A = \frac{1}{\sqrt{4}} \times$$

$$A^T \cdot A = \frac{1}{4} \begin{pmatrix} 4 & -5 & -4 & 5 \\ -5 & 4 & 5 & -4 \end{pmatrix} \begin{pmatrix} 4 & -5 \\ -5 & 4 \\ -4 & 5 \\ 5 & -4 \end{pmatrix} =$$

$$= \frac{1}{4} \begin{pmatrix} 82 & -80 \\ -80 & 82 \end{pmatrix}$$

$$\det\left(A^T \cdot A - \lambda I\right) = 0 \Leftrightarrow \left(\frac{41}{2} - \lambda\right)^2 - (-20)^2 = 0$$

$$\left(\frac{41}{2} - \lambda - 20\right)\left(\frac{41}{2} - \lambda + 20\right) = 0$$

$$\left(\frac{1}{2} - \lambda\right)\left(\frac{81}{2} - \lambda\right) = 0$$

$$\lambda_1 = \frac{81}{2} \qquad \lambda_2 = \frac{1}{2}$$

o $$\left(A^T \cdot A - \lambda_1 I\right) v_1 = 0 \Rightarrow \begin{pmatrix} \frac{41}{2} - \frac{81}{2} & -20 \\ -20 & -\frac{40}{2} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$\begin{cases} -20a - 20b = 0 \\ -20a - 20b = 0 \end{cases} \Rightarrow a = -b$$

$v_1 = (-1 \quad 1)$ normalizare $\Rightarrow$ $v_1 = \left[-\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}}\right]$

$\Rightarrow w_1 = \left[-\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}}\right]$ $\lambda_1 = \frac{81}{2} \cdot \frac{1}{4} = \frac{81}{8}$ prima componentă PCA

- $(A^T \cdot A - \lambda_2 I_2) v_2 = 0$ $\quad \begin{pmatrix} \frac{41}{2} - \frac{1}{2} & -20 \\ -20 & 20 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$

$$\begin{cases} 20a - 20b = 0 \\ -20a + 20b = 0 \end{cases} \Rightarrow a = b$$

$v_2 = [1 \quad 1] \Rightarrow v_2 = \left[\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}}\right]$

$w_2 = \left[\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}}\right]$ $\lambda_2 = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$

3.

4. *Assume we have the design matrix*

$$X = \begin{bmatrix} 2 & -4 \\ -5 & 3 \\ -6 & 2 \\ 1 & -5 \end{bmatrix}.$$

Use PCA to compute the second principal component. Observe that $\boldsymbol{X}$ *is not centered.*

$$\overline{x} = \left(\frac{-3-5}{4} \quad \frac{-1-3}{4} \right) = \left(-2 \quad -1 \right)$$

$$X = \begin{pmatrix} 4 & -3 \\ -3 & 4 \\ -4 & 3 \\ 3 & -4 \end{pmatrix} \qquad A = \frac{1}{\sqrt{4}} X$$

$$A^T \cdot A = \frac{1}{4} \begin{pmatrix} 4 & -3 & -4 & 3 \\ -3 & 4 & 3 & -4 \end{pmatrix} \begin{pmatrix} 4 & -3 \\ -3 & 4 \\ -4 & 3 \\ 3 & -4 \end{pmatrix} =$$

$$= \frac{1}{4} \begin{pmatrix} 50 & -48 \\ -48 & 50 \end{pmatrix} = \begin{pmatrix} \frac{25}{2} & -12 \\ -12 & \frac{25}{2} \end{pmatrix}$$

$$\det\left(A^T \cdot A - \lambda \mathbb{1} \right) = 0 \iff \left(\frac{25}{2} - \lambda \right)^2 - (-12)^2 = 0$$

$$\left(\frac{25}{2} - \lambda - 12 \right) \left(\frac{25}{2} - \lambda + 12 \right) = 0$$

$$\left(\frac{1}{2} - \lambda \right) \left(\frac{49}{2} - \lambda \right) = 0$$

$$\lambda_1 = \frac{49}{2} \qquad \lambda_2 = \frac{1}{2}$$

- $\left(A^T \cdot A - \lambda_1 \mathbb{1} \right) v_1 = 0 \quad \begin{pmatrix} \frac{25}{2} - \frac{49}{2} & -12 \\ -12 & -\frac{24}{2} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$

$\Rightarrow a = -b \qquad v_1 = (-1 \quad 1) \Rightarrow v_1 = w_1 = \left(-\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right) \quad \lambda_1 = \frac{49}{2} \cdot \frac{1}{4}$ → prima comp.

- $\frac{25}{2} - \frac{1}{2} a - 12 b = 0$
  $-12 a + 12 b = 0$

$\Rightarrow a = b \Rightarrow v_2 = w_1 = \left(\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right) \quad \lambda_2 = \frac{1}{8}$

④ ex curs

$$X = \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ -2 & 1 \end{pmatrix} \qquad PCA$$

$m = 3$, $d = 2$

$A = \frac{1}{\sqrt{3}} X$

$A^T \cdot A$, dar dacă am calcula Σ C. Popa a zis să băgăm încă o normalizare de $\frac{1}{m} A^T \cdot A$

$$\Sigma = \frac{1}{3} A^T \cdot A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$\det(A^T \cdot A - \lambda I) = 0$$

$$(2-\lambda)^2 - 1^2 = 0$$

$$(2-\lambda-1)(2-\lambda+1) = 0$$

$$(1-\lambda)(3-\lambda) = 0$$

$$\lambda_1 = 3 \quad \lambda_2 = 1$$

$$A \cdot v = \lambda \cdot v$$

- $$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 3 \cdot \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$\Rightarrow v_1 = -v_2 \Rightarrow v_1 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \quad \lambda_1 = 3 \cdot \frac{1}{3} = 1$$

- la fel și pt $\lambda_2 = 1$

$$\Rightarrow v_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \lambda_2 = 1 \Rightarrow \lambda_2 = 1 \cdot \frac{1}{3} = \frac{1}{3}$$

5.

node gives the smallest entropy.

4. Assume we have the design matrix

$$X = \begin{bmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}.$$

Use PCA to compute the first principal component. Observe that X is not centered.

for two-dimensional points: $\{(2,10),(8,4),(5,8),(6,4),(4,9)\}$. Use Euclidean distance.